Sarvesh Shashidhar
Centre for Machine Intelligence and Data Science
Indian Institute of Technology Bombay

24M2152
M.S. by Research
Gender: Male
DOB: 26/05/2000

Examination	University	Institute	Year	CPI / % Credits
Post Graduation	IIT Bombay	IIT Bombay	2027	8.83 **Core:** 40 **| Total:** 70
Graduation	VTU	RV College of Engineering	2021	8.66
Graduation Specialization: Electronics & Communication Engineering				
Intermediate	CBSE	Jain International Residential School	2017	93.00%
Matriculation	CBSE	Jain International Residential School	2015	10

Papers

- **Sarvesh Shashidhar**, Lankireddy Prabhat, Arpit Agarwal, D. Manjunath, Karan Bhukar, and Tanmay Khandelwal. "Stay or Stray-A Dynamical Systems Viewpoint of Popularity Bias." `arXiv preprint arXiv:2608.10474 (2026)`.
- **Sarvesh Shashidhar**, Ritik, Nachiketa Patil, Suraj Racha, and Ganesh Ramakrishnan. "Inducing Robustness in a 2 Dimensional Direct Preference Optimization Paradigm." `arXiv preprint arXiv:2505.01706 (2025)`.

Research Experience

Modelling Popularity Bias in Recommender Systems | M.S.(R) Thesis | Prof. Arpit Agarwal *(Feb'25 - Present)*
In collaboration with **Amazon Music** team, currently under review at **AAAI 27** (`arXiv pre-print - 2608.10474`).

- Formulated a theoretical framework using **differential equations** modelling popularity bias in recommender systems.
- Derived and proved concrete theorems using stochastic approximation to predict & quantify **user drop-off**.
- Designed a mitigation solution using equalized odds to **prevent popularity bias** without loss in system accuracy.
- Evaluated the framework using **410 million** real-world interactions, validating our predictions and solution.

Professional Experience

AI Research Intern | Motilal Oswal Financial Services Ltd. *(May'26 - Aug'26)*
Technologies: AWS SageMaker, AWS S3, DataBricks, NVIDIA CUDA

- Identified recommendation system biases causing **5%** of 20,000 catalogue items to absorb **84.5%** of interactions.
- Developed **two IPS models** to ensure fairness and integrated them using a 35% signal blend across 7,016 users.
- Achieved **60% reduction** in top 10% popularity concentration, effectively cutting the NDCG@6 cost by half.
- Developed a dynamic **visualisation dashboard** using HTML, CSS, and JavaScript to track model performance.

Associate Member of Technical Staff | athenahealth Technology Pvt. Ltd. *(Jul'21 - Mar'23)*
Technologies: Perforce, JIRA, BitBucket, Kanban workflow

- Engineered clean **Perl rules** code through regression testing, delivering reliable clinical releases for **Rules 1.0** engine.
- Coordinated cross-functional team updates by **resolving JIRA** requests, ensuring sprint timelines were followed.
- Maintained task work orders via **internal Wiki pages**, enabling strict compliance tracking for project leads.

Course Projects

Inducing Robustness in 2D-DPO | Guide: Prof. Ganesh Ramakrishnan | Course: CS 769 *(Mar'25 - May'25)*

- Diagnosed a lack of segment-level **noise robustness** in 2D-DPO, causing a win-rate drop from **43% to 37%**.
- Formulated **Robust 2D-DPO** and trained a Pythia 6.9B model on 6400 HelpSteer-2D prompts to test robustness.
- Attained win-rate rise from **37% to 45%** in the presence of noise, indicating robustness towards noisy labels.
- Accelerated model training and inference by parallelising over **4** $\times$ **NVIDIA H100** GPUs.

Microchess playing agent | Guide: Prof. Shivaram Kalyankrishnan | Course: CS 747 *(Nov'25 - Dec'25)*

- Devised a Microchess agent using **reinforcement learning** to compete against random and rational opponents.
- Engineered Negamax algorithm with Alpha-Beta pruning and **Quiescence search** for modelling gameplay.
- Achieved a **94%** and **84%** win-rate against random and rational opponents respectively across 100 games.

RLHF under User Heterogeneity | Guide: Prof. Arpit Agarwal | Course: CS6103 *(Feb'25 - May'25)*

- Uncovered a flaw in standard RLHF due to the assumption of perfectly homogeneous preferences across all users.
- Engineered **personalised preference models** across 40 workers and 2 distinct clusters using **vector embeddings**.
- Achieved **53% and 52.7%** win-rates, performing better for both user clusters with baseline 52% win-rate.

Fine-tuning & Evaluating CNN Backbones | Guide: Prof. Biplab Bannerjee | Course: GNR638 *(Feb'26 - Mar'26)*

- Bench-marked three **pre-trained CNN** architectures across the aerial image dataset (AID) with **30 classes**.
- Executed 5 evaluation paradigms that highlight few-shot performance and noise robustness across three models.
- Observed **95% accuracy** and **24% relative drop** due to noise for ConvNeXt-Tiny (**27.8 million** parameters).

Blackjack Bot | Guide: Prof. Shivaram Kalyankrishnan | Course: CS 747 *(Oct'25 - Nov'25)*
- Modelled a 26-card Blackjack variation as a **Markov Decision Process** with three specific actions.
- Implemented Linear Programming and **Howard's Policy Iteration** to solve linear constraints across scenarios.
- Evaluated optimal value functions against agent actions and successfully passed **100%** of defined test cases.

OCR-based MCQ solver | Guide: Prof. Biplab Bannerjee | Course: GNR638 *(Apr'26 - May'26)*
- Procured and scraped high-quality MCQ questions on topics pertaining to **Deep Learning** to train an OCR solver.
- Trained a **Qwen3-VL** model with **8 billion** parameters to perform OCR extraction and solve MCQ questions.
- Attained **90% accuracy** using few-shot training when graded against pre-defined evaluation dataset.

Emotion Classification for Hindi | Guide: Prof. Pushpak Bhattacharya | Course: CS 626 *(Oct'24 - Dec'24)*
- Classified **20 thousand** sentences in Hindi into five distinct emotion classes using deep learning models.
- Verified **162 scenarios** – 9 deep learning models across 6 dataset variations using 3 word-embedding techniques.
- Attained **84.3% accuracy** using TextBlob embeddings and mBERT model, with **human-annotated labels**.

Improved ResNet Implementation | Guide: Prof. Biplab Bannerjee | Course: GNR638 *(Mar'26 - Apr'26)*
- Addressed information flow obstruction, lossy projection and **sub-optimal normalisation** of deep ResNet-50.
- Implemented **iResNet-50** model from scratch and trained for 100 epochs to overcome the bottlenecks of ResNet-50.
- Achieved an **85.31% Top-1 accuracy** using a **25.56 million** parameter iResNet-50 on CIFAR-10 & CIFAR-100.

Other Projects

Machine Learning over Encrypted Data | Self Project *(2020 - 21)*
- Identified **250 million** medical data breaches and the need for encryptions when dealing with medical data.
- Developed a Python framework using **Homomorphic Encryption**, enabling operations directly on encrypted data.
- Performed linear & logistic regression accurately on encrypted data **without decryption**.

Positions of responsibility

Department General Secretary | C-MInDS, IIT Bombay *(Jun'25 - Apr'26)*
- Organised **India AI Impact Summit 2026** pre-event and **TechConnect 2025** fostering industry collaborations.
- Created a comprehensive **placement pitch deck** and actively pushed for student and industry interactions.
- Launched a student **grievance redressal portal** and helped modify **course buckets** as per students' demands.
- Successfully hosted the annual **orientation** and **convocation** ceremonies for freshers and graduating students.

Academic Council Coordinator | Post Graduate Academic Council, IIT Bombay *(Jan'25 - Jun'25)*
- Moderated **Teachers for Tomorrow 2.0** panel to give students an insight into the journey of young professors.
- Assisted in organising pre-placement events and industry talks as a part of the **events vertical**.

Head Teaching Assistant | IITB-KEF AI-ML Course | Prof. Vinay Kulkarni, Prof. Mani Bhushan *(Mar'26 - Aug'26)*
- Coordinated operations and classes for **100 students** as part of Kotak Education Foundation.
- Organised **30 hours** of classes while creating & **scoring course assignments** and maintaining attendance records.

Teaching Assistant | Introduction to Machine Learning (DS 303) | Prof. Parthe Pandit *(Jan'26 - May'26)*
- Graded assignments & examinations, and provided **academic support** to students via targeted TA sessions.

Teaching Assistant | Programming for Data Science (DS 203) | Prof. Vinay Kulkarni *(Aug'25 - Dec'25)*
- Assessed student performance, resolved queries & provided **actionable feedback** during TA meetings.

Teaching Assistant | Applied Data Science and Machine Learning (ME 228) | Prof. Alankar *(Jan'25-May'25)*
- Designed tutorials covering **time series** data analysis, evaluated exams & projects, and clarified student queries.

Relevant Certifications

- **Data Analysis and Visualisation with Power BI** | Microsoft (Coursera) *(2024)*
- **Introduction to Big Data** | UC San Diego (Coursera) *(2021)*
- **Git workshop** | IEEE Computer Society, RVCE Student Chapter *(2020)*
- **Introduction to Data Analytics for Business** | University of Colorado, Boulder (Coursera) *(2020)*
- **Machine Learning with AI using Python** | Knowledge Solutions India *(2020)*

Relevant Courses Undertaken

- Foundations of Machine Learning
- Speech and Natural Language Processing and the Web
- Foundations of Intelligent and Learning Agents
- Human-Centered AI: From Learning Models to Responsible Systems
- Theoretical Machine Learning
- Optimisation in Machine Learning
- Algorithms and Complexity
- Online Learning and Optimisation

Technical Skills

- **Programming & Scripting Languages**: Python, C++, SQL, Bash, LaTeX
- **ML tools & libraries**: PyTorch, Scikit-Learn, JAX, HuggingFace, Jupyter, Anaconda, Pandas, NumPy, SciPy, Matplotlib, Seaborn.
- **HPC & Server operations**: CUDA, Linux, SLURM, AWS SageMaker.
- **Project Management**: Git, WANDB, AWS S3, Docker, Excel, PowerBI.
- **Recommendation Systems tools**: RecBole, Elliot, Cornac.

Extracurricular Activities

- Secured **World Rank 13** in **IMC Prosperity 3** by creating quant models to capitalise on market trends. *(Feb'25)*
- Participated as an **instructor** at the two-day **Hands-on Workshop** organised by KCDH, IIT Bombay. *(Mar'25)*
- Secured an **International Win** at Formula Hybrid as part of Team Ashwa Racing. *(2020)*